Uncertainty-Aware Relative Compliance Estimation of Wind-Excited Plant from Monocular Video

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***Abstract*—Quantifying the mechanical properties of plant branches before manipulation can be valuable for automated and selective harvesting of fruits and vegetables. This is fundamentally challenging because estimation of properties such as absolute stiffness requires known force measurements, which are unreliable, expensive, and time-consuming to collect at scale. We identified artificial wind excitation as an acceptable proxy, a *relative projected compliance* κ_e that measures how much each branch segment bends compared to others under shared excitation, without requiring force measurements. Our pipeline combines an initial point tracking using TAPIR, a factor-graph-style trajectory smoother, and a separable low-rank compliance model fitted by alternating least squares. We measured the stability of the fitted κ_e, which gives us a relative uncertainty proxy for the estimates. Using a monocular video of a wind-excited plant branch with 20 annotated nodes, 19 edges, and 1104 frames, we demonstrate our proof of concept. Our pipeline recovers a physically plausible proximal-to-distal relative motion gradient and an interpretable relative compliance map with 95% confidence interval. These results demonstrate that relative compliance maps with corresponding uncertainty are recoverable from monocular video without ground-truth force data, enabling new possibilities for agricultural robotic branch manipulation planning.**

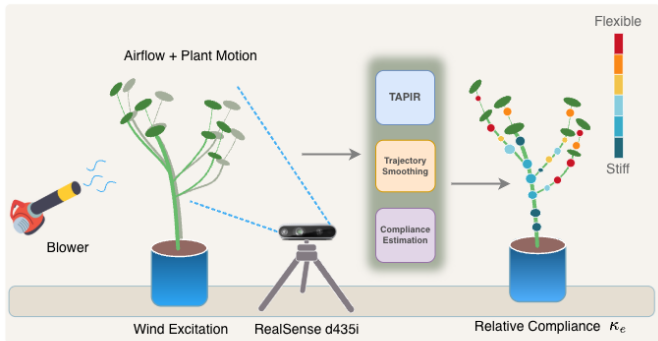

Fig. 1: In our current pipeline, the monocular video input of a wind-excited plant is processed through TAPIR-based [2] tracking and trajectory smoothing, which computes and produces the image-plane relative compliance of the plant stem at various nodes.

I. Introduction

Understanding the mechanical flexibility of plant branches is a key capability for agricultural robots involved in selective harvesting and manipulation within cluttered canopies. In practice, a robot must predict how a branch will deform under contact forces before manipulation. Yet acquiring this information at scale is difficult: contact-based stiffness measurement requires a physical probe. Therefore, wind force is a reasonable proxy that can be produced by an artificial source such as a blower or pressurized air. This paper aims to use this proxy wind force combined with vision-based approaches to estimate relative flexibility for branch segments affected by the wind load.

Prior work on plant motion analysis has largely focused on tracking or segmentation of plant structure [1], rather than recovering material properties from dynamic video. Work on flexible object state estimation and compliant robot manipulation has addressed related problems in controlled settings, but has not addressed the open-world case where loading is unknown and non-uniform.

In this paper, we make the following contributions:

1) We introduce the relative compliance proxy κ_e from a monocular video of a branch under unknown wind load. This κ_e is only the image-plane projected component of the full 3D compliance.
2) We implemented a trajectory smoother using factor-graph based optimization, which enforces temporal smoothing and inextensibility of edges while respecting the TAPIR tracking.
3) We provide uncertainty estimates for κ_e via moving-block bootstrap, appropriate for temporally correlated bending signals.
4) We projected an interpretable compliance gradient on a real-time video sequence and its corresponding 3D point cloud.

The further structure of this paper is as follows: Section II contains the problem formulation, which discusses the identifiability of relative compliance. Section III includes the proposed 3-stage methodology for compliance estimation and uncertainty calculation. Additionally, Section IV explains the experiment setup and discusses the results achieved. Lastly, V concludes the paper with the future implications of the proposed method.

II. Problem Formulation

We model each branch segment as an Euler-Bernoulli beam [5]. Let $s \in [0, L]$ denote arc-length along the branch segment,

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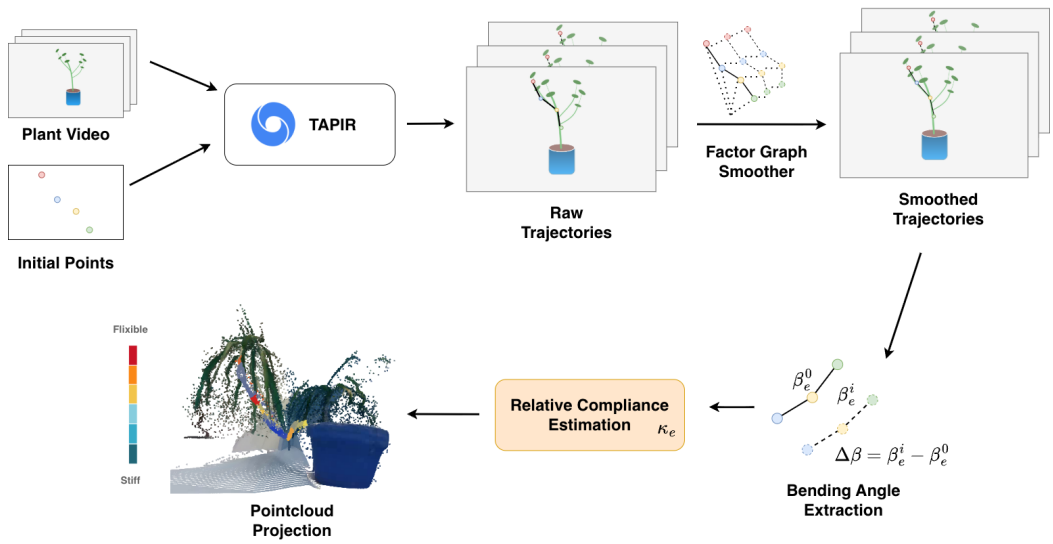


Fig. 2: Overall Methodology for projected (2D) relative compliance estimation. We start with a video of a plant branch and corresponding annotated tracked points, which are processed through TAPIR, and the raw trajectories are generated. We use factor graph style smoothing to recover smooth and noise-free trajectories from the moving average of the raw trajectories. Finally, we extract bending angles and estimate relative compliance through alternating least squares [3] and project the compliance heatmap to a 3D point cloud generated from the estimated Monocular depth by DepthAnything V2 [4].

with $s = 0$ near the parent joint and $s = L$ at the distal tip. Let $v(s,t)$ be the transverse deflection at time t, and let $q(s,t)$ be the distributed transverse load induced by the airflow.

The standard Euler-Bernoulli equation [5] is

$$EI\frac{\partial^4 v(s,t)}{\partial s^4} = q(s,t), \tag{1}$$

where EI is the bending stiffness, with E the Young's modulus and I the second moment of area. Equation (1) states that the applied load is proportional to the fourth derivative of deflection.

We instead express the equation in terms of the bending moment. By integrating twice and applying Cauchy's formula [6] for repeated integrals, we get

$$EI\frac{\partial^2 v(s,t)}{\partial s^2} = \int_s^L q(\xi,t)(\xi - s)\,\mathrm{d}\xi. \tag{2}$$

This second-order integral form is equivalent to the standard fourth-order Euler-Bernoulli equation [5]. However, it explicitly shows how distal airflow loads generate bending moment through their lever arms.

A. Why Absolute Stiffness Is Not Identifiable

Equation (2) can be rearranged as

$$\frac{\partial^2 v(s,t)}{\partial s^2} = \frac{1}{EI}\int_s^L q(\xi,t)(\xi - s)\,\mathrm{d}\xi. \tag{3}$$

Thus, the observable curvature is proportional to the applied load and inversely proportional to bending stiffness.

From monocular video, we observe the branch motion, from which we estimate a discrete bending signal. However, the airflow load $q(s,t)$ is not measured. This creates a fundamental scale ambiguity. For any positive scalar c,

$$EI \mapsto c\,EI, \qquad q(s,t) \mapsto c\,q(s,t) \tag{4}$$

leaves Eq. (3) unchanged, because both numerator and denominator are scaled by the same factor. Therefore, the same observed motion can be explained either by a stiff branch under a strong load or by a compliant branch under a weak load.

Consequently, absolute bending stiffness EI is not identifiable from monocular motion alone unless the external load is known or independently measured.

B. What Is Identifiable: Image-Plane Projected Relative Compliance

Although absolute stiffness cannot be recovered, projected relative compliance between branch segments can be estimated under a shared-excitation assumption,i.e., wind force is constant throughout the branch at any time instance. In our setting, the branch is excited by a common airflow source, such as a blower or pressurized air, in a constant direction throughout the estimation period. Therefore, we can assume that the wind load is approximately distributed across segments even if its intensity changes with time.

We model the load on edge e as separable into a temporal excitation amplitude and a spatial load profile:

$$q_e(\xi,t) \approx a_t\,\bar{q}_e(\xi), \tag{5}$$

where $a_t \geq 0$ captures the airflow intensity at frame t, and $\bar{q}_e(\xi)$ captures the spatial distribution of load along edge e.

Substituting Eq. (5) into Eq. (2) gives

$$EI_e \frac{\partial^2 v_e(s,t)}{\partial s^2} = a_t \int_s^{L_e} \bar{q}_e(\xi)(\xi - s)\,\mathrm{d}\xi. \tag{6}$$

Rearranging,

$$\frac{\partial^2 v_e(s,t)}{\partial s^2} = a_t \frac{1}{EI_e} \int_s^{L_e} \bar{q}_e(\xi)(\xi - s)\,\mathrm{d}\xi. \tag{7}$$

From the video, we do not directly measure continuous curvature. Instead, for each edge e and frame t, we compute a discrete bending observation $y_{e,t}$. It's the change in local angle between tracked skeleton points. This quantity is a finite-dimensional proxy for the curvature term in Eq. (7). Applying this discrete measurement operation to Eq. (7) yields

$$y_{e,t} \approx a_t \underbrace{\left[\frac{1}{EI_e} C_e\right]}_{\text{edge-dependent compliance}}, \tag{8}$$

where C_e absorbs the edge-specific geometric and load-shape terms induced by the integral and by the discrete angle measurement.

We therefore define an effective relative compliance proxy of edge e as

$$\kappa_e \propto \frac{C_e}{EI_e}, \tag{9}$$

and define a shared frame-wise excitation proxy z_t proportional to the airflow amplitude a_t. This gives the rank-1 observation model

$$y_{e,t} \approx \kappa_e z_t. \tag{10}$$

Here, $y_{e,t}$ is the observed bending signal for edge e at frame t, $\kappa_e > 0$ estimates the relative projected compliance of that edge, and $z_t \geq 0$ is the shared excitation magnitude at frame t. Edges with $\kappa_e > 1$ bend more than average under the same excitation and are therefore relatively compliant. Edges with $\kappa_e < 1$ bend less than average and are therefore relatively stiff.

III. Methodology

The full pipeline has three stages: (1) trajectory smoothing via factor-graph optimisation; (2) bending angle extraction with outlier rejection; (3) relative projected compliance estimation with bootstrap uncertainty.

A. TAPIR Point Tracking

We use TAPIR (Tracking Any Point with per-frame Initialization and temporal Refinement) to track the movement of all the annotated points or nodes of the plant stem under wind load. TAPIR was introduced by Doersch et al. [2], which takes a set of initialized query points on visible plant structures and predicts their corresponding image locations in subsequent frames. It also estimates whether each point is visible or occluded. TAPIR uses a 2-stage framework:

- a per-frame matching system which independently proposes potential trackpoints at each timeframe.
- in next stage uses temporal context and local correspondence to refine the proposed trajectories

This makes TAPIR suitable for tracking points on deforming objects such as plant branches, where reliable frame-to-frame correspondence is required despite motion, appearance changes, and temporary occlusions. However, these raw tracks produced by TAPIR are prone to disturbances due to pixel noise and changes in lighting. To smooth these unwanted jitters, we used a factor-graph-based smoother.

B. Factor-Graph Trajectory Smoother

After extracting the raw trajectories from TAPIR, we first apply a simple moving average and then use Factor-Graph style optimization [7] to reduce the noise and smooth trajectories. Let $\boldsymbol{x}_{i,t} \in \mathbb{R}^2$ be the latent smoothed 2-D position of node i at frame t, and let $\boldsymbol{y}_{i,t}$ be the corresponding raw inputs. The branch is represented as a rooted tree with node set $\mathcal{V}$ and edge set $\mathcal{E}$. We minimise a quadratic energy:

$$\begin{aligned}\mathcal{L}(\boldsymbol{X}) = \lambda_m &\sum_{i,t} \|\boldsymbol{x}_{i,t} - \boldsymbol{y}_{i,t}\|^2 \\ &+ \lambda_v \sum_{i,t} \|\boldsymbol{x}_{i,t+1} - \boldsymbol{x}_{i,t}\|^2 \\ &+ \lambda_a \sum_{i,t} \|\boldsymbol{x}_{i,t+1} - 2\boldsymbol{x}_{i,t} + \boldsymbol{x}_{i,t-1}\|^2 \\ &+ \lambda_e \sum_{(p,c)\in\mathcal{E},t} \left(\|\boldsymbol{x}_{c,t} - \boldsymbol{x}_{p,t}\| - l_{pc}^0\right)^2\end{aligned} \tag{11}$$

where $l_{pc}^0 = \|\boldsymbol{r}_{pc}^0\|$ is the initial edge length (encoding approximate inextensibility), and $\lambda_m, \lambda_v, \lambda_a, \lambda_e$ balance measurement fidelity against physical priors.

The first term, λ_m, is measurement fidelity, which prevents the smoothed tracks from deviating too far from the actual visual data seen by TAPIR. The 2nd and 3rd terms (λ_v, λ_a,) are for velocity and acceleration smoothness, respectively. They penalize any sudden change in position or velocity and encourage a steady flow of motion. The last term λ_e enforces structural constraints, so that the branch segments (edges) should maintain their original length even as they bend. By framing the problem this way, we can enforce physical consistency as it smooths not only each point individually but also the entire skeleton as one connected object. Since (11) is a quadratic in $\boldsymbol{X}$, the global minimum is found by solving a sparse linear system, which is efficient even for $\sim$20 nodes over $\sim$1000 frames.

C. Bending Angle Extraction

For each ordered triplet (grandparent g, parent p, child c) with $e = (g, j, c)$, the instantaneous bending angle at frame t is:

$$\beta_{e,t} = \arccos\left(\frac{\boldsymbol{v}_{gp,t} \cdot \boldsymbol{v}_{pc,t}}{\|\boldsymbol{v}_{gp,t}\|\,\|\boldsymbol{v}_{pc,t}\|}\right) \tag{12}$$

where $\boldsymbol{v}_{gp,t} = \boldsymbol{x}_{g,t} - \boldsymbol{x}_{p,t}$ and $\boldsymbol{v}_{pc,t} = \boldsymbol{x}_{c,t} - \boldsymbol{x}_{p,t}$. The initial angle β_e^0 is taken from the first frame without any excitation. The signed deviation $\Delta\beta_{e,t} = \beta_{e,t} - \beta_e^0$ is then computed for each frame. The outliers are rejected via a median absolute

deviation (MAD) mask [8]. The bending signal described in our rank 1 observation model (10) is $y_{e,t}$, which is infinitesimal in nature. Here, we use the $|\Delta\beta_{e,t}|$ as a finite and discrete proxy of the observation signal. In other words, the signal entering the compliance model is $y_{e,t} = |\Delta\beta_{e,t}|$ for all inlier frames.

D. Relative Compliance Estimation

1) Alternating least squares: Given bending signals $\{y_{e,t}\}$, we estimate the image-plane projected relative compliance (κ_e) and the wind-excitation magnitude at time t (z_t) by alternating:

Initialisation: $z_t^{(0)} = \mathrm{median}_e\, y_{e,t}$.

Update κ_e given $\boldsymbol{z}$: The raw least-squares slope is

$$\hat{\kappa}_e^{\mathrm{raw}} = \frac{\sum_t z_t\, y_{e,t}}{\sum_t z_t^2} \tag{13}$$

However, just using alternating least squares [3] has a major problem. During the video, some parts of the branch may be occluded behind leaves for long periods. If an edge is only seen for a few frames, a standard least-squares ($\hat{\kappa}_e^{raw}$) might produce a noisy estimate because it doesn't have enough evidence to form a stable estimate. We use shrinkage toward neutral as a regularization method. Where neutral compliance is $\kappa = 1$, i.e., a segment with $\kappa_e = 1$ bends exactly as much as the branch average under a given wind load. In practice, we implement shrinkage toward neutral in log space:

$$\kappa_e = \exp\!\big(\alpha_e \log \hat{\kappa}_e^{\mathrm{raw}}\big), \quad \alpha_e = \frac{n_e}{n_e + \lambda} \tag{14}$$

where n_e is the number of inlier frames for edge e and λ is a prior strength hyperparameter. Equation (14) implements a log-normal shrinkage prior: edges with few observations are pulled toward $\kappa_e = 1$ (neutral compliance).

Update z_t given $\boldsymbol{\kappa}$: $z_t = \big(\sum_e \kappa_e\, y_{e,t}\big)/\big(\sum_e \kappa_e^2\big)$.

After each iteration, the geometric mean of $\{\kappa_e\}$ is renormalised to unity to resolve scale ambiguity.

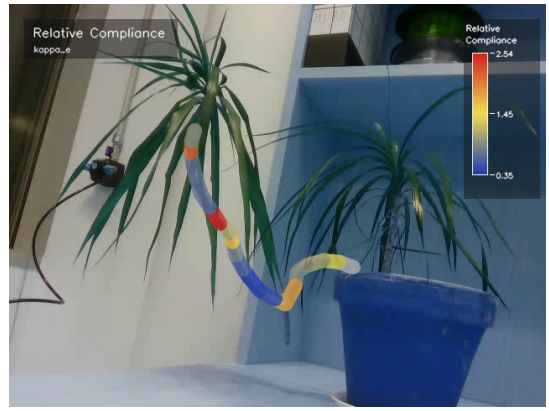
2) Uncertainty: moving-block bootstrap.: Because bending time series are temporally autocorrelated, the standard iid bootstrap underestimates uncertainty. To avoid it, we use a moving-block bootstrap [9]. Instead of picking single frames, it picks contiguous blocks of time B, which are resampled with replacement to form synthetic sequences of length T. Then the full compliance estimation model is re-fitted on each. The 95% confidence interval for κ_e is the 2.5th-97.5th percentile of the bootstrap distribution $\{\kappa_e^{(b)}\}$.

The relative Confidence Interval width $U_e = (CI_{\mathrm{hi},e} - CI_{\mathrm{lo},e})/\kappa_e$ serves as a scale-normalised uncertainty index. A value of $U_e < 0.2$ indicates well-constrained estimates; $U_e > 0.5$ flags unreliable edges.

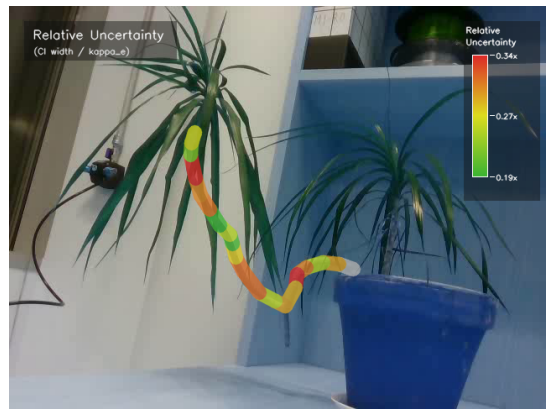
IV. Experiments and Results

A. Experimental setup

A potted plant with long branches was placed in front of an Intel RealSense D435i camera on a tripod. The plant is excited using a constant velocity air source. The camera was mounted in a fixed position to avoid background motion, and we used only the monocular RGB video stream as input.



((a)) Relative Projected Compliance of the stem, where blue indicates the stiff sections and red indicates the more compliant sections.



((b)) Relative Uncertainty of the stem indicating the measurement uncertainty of κ_e. Here, green shows more confidence in the compliance estimation, and red indicates more uncertainty.

Fig. 3: Projection of estimated κ_e and corresponding uncertainty as a heatmap in the Image frame.

Initial anchor points on the branch were annotated manually for the first frame. The movement of these points under air excitation is tracked through 1104 frames using TAPIR by Doersch et al. [2]. The branch was represented as a chain of 20 nodes with 19 edges. This linear representation remained fixed throughout the video, once it was defined for the first frame skeleton.

The tree is traversed from the root node to the tip node, with the root distance increasing monotonically from edge 1 to edge 19. The edges are assigned to three anatomical zones based on their distance from the root: proximal (the first 0-5 edges), mid (edges 6-12), and distal(last 7 edges (edges 13-19)).

The raw trajectories produced by TAPIR are first processed with a moving average filter using a window size of 5, and are then smoothed via factor-graph style optimization with the following hyperparameters: $\lambda_m = 1.0$, $\lambda_v = 0.35$, $\lambda_a = 0.2$, $\lambda_e = 0.45$.

Next, we estimated the relative compliance with Alternating Least Squares [3] and Uncertainty with the moving-block bootstrap with the following hyperparameters: Prior strength

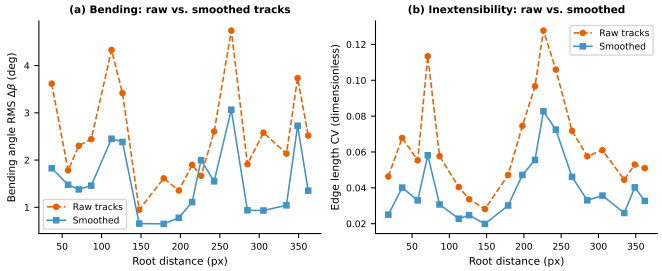


Fig. 4: Effect of factor-graph smoothing. Left: bending angle RMS across the branch. Right: edge-length coefficient of variation. Smoothed tracks (blue) show ∼39% lower noise than raw TAPIR tracks with just a moving average filter (orange) while preserving the motion signal.

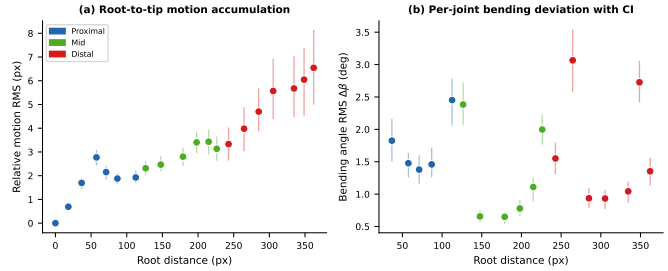


Fig. 5: Branch motion analysis. (a) Node relative motion RMS as a function of root distance, showing monotonic amplification toward tips. (b) Per-joint bending angle RMS $\Delta\beta$ with 95% CIs (vertical bars).

$\lambda = 12$; block bootstrap with block size $B = 25$ frames; 250 bootstrap repetitions.

B. Impact of Smoothing in Noise Reduction

The factor-graph based smoother substantially reduces the track's geometric noise while preserving the bending signal (Fig. 4). We used the Root Mean Square(RMS) of the bending signal ($y_{e,t}$) as it effectively accounts for both the positive and negative bending of the stem from its resting position over the entire 1104-frame video and boils it down into a single intensity score. Across all the 18 bends (leaving the root node), the mean bending RMS decreases from 2.53° (raw moving average) to 1.54° (smoothed), which is a 39.1% reduction. We used the edge length coefficient of variation

$$CV = \frac{\sigma}{\mu}$$

to measure the relative variations of the branch's edge length across the frames, where the σ is the standard deviation and μ is the mean of edge lengths. The CV decreases from 0.065 to 0.040 (38.7% reduction) after smoothing, confirming that smoothed tracks better respect the inextensibility prior. The reduction is consistent across all zones.

C. Motion Amplification Along the Branch

Since we modeled our branch segments as an Euler-Bernoulli Beam [5], the motion is always monotonically increasing toward the tip because of the lever arm effect. Fig. 5(a) shows node relative motion RMS increasing almost monotonically from the root to the distal tip. This root-to-tip amplification is consistent with the cantilever beam prediction. On the other hand, Fig. 5(b) shows bending angle deviations per ordered triplets. Several bends show elevated RMS ($> 2°$) while others are near zero, reflecting heterogeneous compliance along the branch and not merely as a function of distance from the root.

D. Relative Projected Compliance Map

Fig. 6(a) shows κ_e with 95% bootstrap confidence intervals (CIs) for all 18 edges. The root edge (edge 1) is not considered since it has zero compliance, and it can only be a grandparent node in any ordered triplet.

The highest compliance is observed at edge 14 (distal zone) with $\kappa_e = 2.54$ and 3.07° deviation angle as shown in the plot 6(a) and 5(b) respectively. Other highly compliant edges are edge 6 (proximal zone), edges 7 and 12 in the mid zones, and edge 18 in the distal zone. This observation anatomically corresponds to Image 3 (a), where these compliant edges are marked with red, yellow, and orange overlays. However, most of the edges are not as compliant, and we can clearly observe 2 different, relatively stiff branch sections in the mid and distal regions. It corresponds to edges (8-11) and (15-17) in Figure 6 and blue overlays in the figure 3 (a).

The recovered compliance pattern is physically interpretable. Since the branches are not uniform and they have bends and curves, which can act as a spring under sufficient wind excitation. We observe (Fig.3(a)) that the highly compliant areas correspond to the 3-4 bends in the stem of the plant. However, the least compliant areas are usually the stiff and straight parts of the branch, which are more resistant to wind forces because of their geometry. It also matches our own physical world models, and can be useful information that we can transmit to Harvester robots.

The non-monotonic compliance-vs-depth profile is important as it shows that the method captures spatially varying compliance rather than simply amplifying motion with distance. This distinguishes κ_e from a pure motion amplitude measure.

E. Uncertainty Analysis

Fig. 6(b) shows the relative Confidence Interval width U_e for each edge. The median is 0.29 (29%), indicating well-constrained estimates for most edges. Edges with $U_e > 0.4$ tend to be in the distal or mid zones. The distal region has shorter, fewer inlier frames due to repeated occlusions by the leaves. On the other hand, the mid zone has low motion signals, so it's hard to differentiate actual signals from pure noise. Here we can also observe the importance of regularization, the shrinkage prior (14) pulls weakly-supported edges toward $\kappa_e = 1$ and their bootstrap Confidence Intervals widen accordingly.

F. Shared Excitation Proxy

Fig. 7 shows the estimated shared excitation z_t over the full sequence. This result is physically consistent with the video

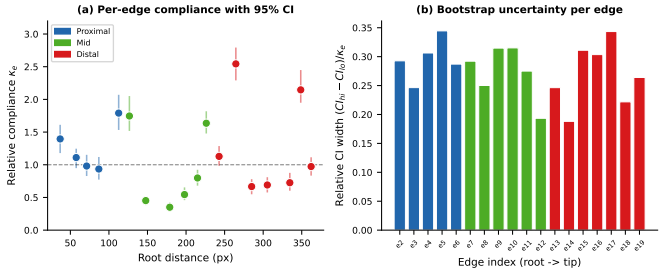


Fig. 6: Image plane projected Relative compliance estimates. (a) κ_e per edge as a function of root distance; vertical bars are 95% moving-block bootstrap CIs; dashed line marks neutral compliance ($\kappa = 1$). Colours indicate anatomical zone. (b) Bootstrap uncertainty (relative CI width) per edge; edges with poor support show higher uncertainty.

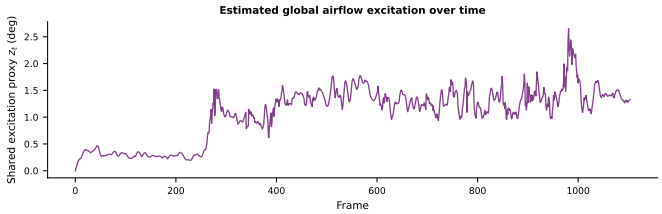


Fig. 7: Estimated shared excitation proxy z_t over the video sequence (1104 frames), capturing periodic airflow from the blower.

observation, where we can see, initially for approximately 8 seconds, there was no wind, and the first wind gust arrived at around the 264th frame. The signal also exhibits structured oscillations consistent with periodic wind gusts from the blower, with an RMS of 1.18°. The temporal structure of the excitation z_t supports the shared-excitation assumption as well. If different edges had unrelated excitation patterns, the alternating optimisation would not converge to a consistent z_t.

V. Conclusion and Future Works

We presented a methodology to estimate the relative projected compliance between branch segments using a monocular video feed of a plant under wind load. Our method doesn't require ground truth for the applied force, as our predictions are relative in nature. The key insight is that, using the rank-1 observation model, we can estimate both compliance and shared temporal excitation for every edge. These estimations are physically consistent with the geometry of the plant and its video observation data. To enforce the physical priors of the plant, like inextensibility and temporal smoothness, we used factor-graph style optimisation. This reduced the geometric noise by 39% over raw tracks produced by TAPIR [2] with only moving average smoothing. We measured the uncertainty of our estimations using the moving-block bootstrap method, which produced a median relative confidence interval of 29%. Finally, the method is applied over a 20-node branch graph through 1104 frames to produce an interpretable compliance map. This work will contribute towards the careful harvesting of produce based on plant's stiffness. The method can estimate the compliance of the stem without touching it, contributing to the safe manipulation of plants by the robot.

A. Limitations and Future Work

The current results are from a single plant, with only once wind source and captured in a single session. Multiple configurations are required to confirm the stability of the relative compliance estimation. Furthermore, a crucial assumption made in this paper is that the amplitude of wind force is consistent throughout the branch at any time instance. However, in the real world, the wind load is proportional to the surface area (leaves attached to the plant) even with consistent air flow. Moreover, future work should integrate automatic anchor point detection from segmented plant branches by using methods like SAM2 [10]. This automated initialization will allow for estimating bootstrap uncertainties over topological variants of the initial skeleton. Lastly, our model expects the movement induced by the wind excitation to be planar in nature and parallel to the image plane. For any non-planar movements introduced by the wind load that cannot be observed by a single monocular camera, we need multiview cameras and updated methods.

Acknowledgments

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References

[1] M. Minervini, H. Scharr, and S. A. Tsaftaris, "Image analysis: the new bottleneck in plant phenotyping," *IEEE Signal Processing Magazine*, vol. 32, no. 4, pp. 126–131, 2015.

[2] C. Doersch, Y. Yang, M. Vecerik, D. Gokay, A. Gupta, Y. Aytar, J. Carreira, and A. Zisserman, "TAPIR: Tracking any point with per-frame initialization and temporal refinement," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 10 061–10 072.

[3] T. Hastie, R. Mazumder, J. D. Lee, and R. Zadeh, "Matrix completion and low-rank svd via fast alternating least squares," *The Journal of Machine Learning Research*, vol. 16, no. 1, pp. 3367–3402, 2015.

[4] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth anything v2," *Advances in Neural Information Processing Systems*, vol. 37, pp. 21 875–21 911, 2024.

[5] J. M. Gere and B. J. Goodno, *Mechanics of Materials*, 7th ed. Stamford, CT: Cengage Learning, 2009.

[6] G. B. Folland, *Advanced Calculus*. Upper Saddle River, NJ: Prentice Hall, 2002.

[7] M. Xie, A. Escontrela, and F. Dellaert, "A factor-graph approach for optimization problems with dynamics constraints," *arXiv preprint arXiv:2011.06194*, 2020.

[8] P. Rousseeuw and C. Croux, "Alternatives to median absolute deviation," *Journal of the American Statistical Association*, vol. 88, pp. 1273 – 1283, 12 1993.

[9] P. Bühlmann and H. R. Künsch, "Block length selection in the bootstrap for time series," *Computational Statistics & Data Analysis*, vol. 31, no. 3, pp. 295–310, 1999.

[10] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson *et al.*, "SAM 2: Segment anything in images and videos," in *arXiv preprint arXiv:2408.00714*, 2024.